

SDN Security Analysis System

Network Topology Report

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Executive Summary

This report covers the security assessment of the SDN infrastructure. A total of 0 vulnerabilities were identified, of which 0 are critical severity and 0 have been patched. 1 attack simulations were executed, with 1 demonstrating successful exploitation paths.

Network Infrastructure Overview

Component	Name	IP Address	Status
Controller	AES-256	192.168.12	ACTIVE

Security Recommendations

Priority	Recommendation
Critical	Enable TLS mutual authentication for all controller-switch OpenFlow connections
Critical	Implement dynamic ARP inspection to prevent ARP spoofing attacks
High	Deploy rate limiting on control plane to prevent DoS/saturation attacks
High	Enable controller clustering to eliminate single points of failure
High	Implement cryptographic signing of flow rules to prevent tampering
Medium	Deploy network anomaly detection at the SDN controller
Medium	Regularly audit flow tables for unauthorized or conflicting rules
Medium	Separate management plane from data plane networks
Low	Implement comprehensive logging and SIEM integration
Low	Conduct periodic formal verification runs to confirm security properties